Naming Organic Compounds

Exam code: 0620 & 0971

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Naming organic compounds

- The names of organic compounds have two main parts:
  - the stem (sometimes called the prefix)
  - end part (or suffix)
- The stem indicates the number of carbon atoms present in the longest continuous chain of the compound

Number of carbon atoms in the longest chain	Part of the chemical's name
1	meth
2	eth
3	prop
4	but
5	pent
6	hex

- The end part of the name tells you what **functional group** is in the compound

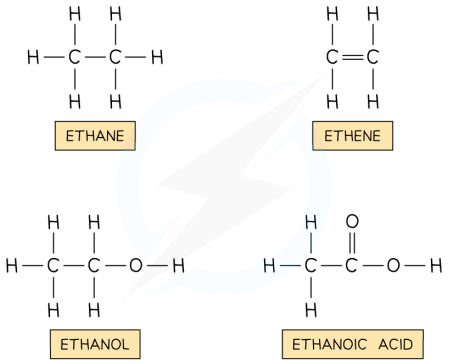
End part of the name	Functional group	Organic family
ane	none (contains only C-C bonds)	Alkane
ene	C=C bond	Alkene
anol	-OH	Alcohol

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

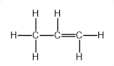
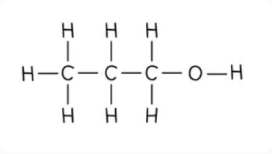
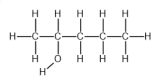
amine	$-NH_2$	Amine
-yl -anoate	-COO-	Ester

Structures of organic compounds



Worked Example

Name the following organic compounds:

1	2	3
		

Answers:

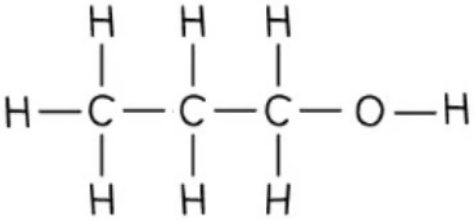
1. Propene
   - The longest carbon chain is 3 carbons, so the name contains **prop**
   - The functional group is C=C, so the name contains **-ene**
2. Propanol
   a. The longest chain is 3 carbons, so the name is **prop**
   b. The functional group is OH, so the name contains **-anol**
3. Pentanol
   - The longest carbon chain is 5 carbons, so the name contains **pent**
   - The functional group is OH, so the name contains **-anol**

Examiner Tips and Tricks

Further naming of organic compounds

Extended tier only

- When there is more than one carbon atom where a functional group can be located it is important to distinguish exactly **which carbon** the functional group is on
- Each carbon is numbered and these numbers are used to describe where the functional group is
- For example:
  - Propan-1-ol is an alcohol with an -OH functional group
  - The 1 in the name indicates that the -OH group is located on the first carbon atom
  - **Careful:** There are many times when the numbering should start from the right - this is to keep the numbers as low as possible in chemical names



In propan-1-ol the -OH group is located on the first carbon atom

Alkanes

- As before, the number of carbon atoms gives the part of the compounds name
- Alkanes contain only **single carbon-carbon bonds**, **C-C**, which means that their name ends with **-ane**
  - Some compounds, like alcohols and carboxylic acids, have -an- in their name
  - This indicates that those compounds contain only single carbon-carbon bonds, C-C

Alkenes

- As before, the number of carbon atoms gives the part of the compounds name
- Alkenes contain at least one **double carbon-carbon bond**, **C=C**, which means that their name ends with **-ene**
  - The first alkene is ethene because you must have two carbons to be able to form a double carbon-carbon bond, C=C
- After propene, you must state the number of the first carbon that is part of the double carbon-carbon bond,

- But-2-ene has a double carbon-carbon bond, C=C, on the second carbon in the chain

Alcohols

- As before, the number of carbon atoms gives the part of the compounds name
- Alcohols contain an **alcohol / hydroxyl group**, **–O–H**, which means that their name ends with **–anol**
- After ethanol, you must state the number of the carbon that has the alcohol / hydroxyl group, –O–H, attached
  - e.g. propan-1-ol has the alcohol / hydroxyl group, –O–H, on the first carbon in the chain
    Propan-2-ol has the alcohol / hydroxyl group, –O–H, on the second carbon in the chain

- As before, the number of carbon atoms gives the part of the compounds name
- Carboxylic acids contain a **carboxylic acid group**, **–COOH**, which means that their name ends with **–anoic acid**
- There is no need to number carboxylic acids because the carbon that is part of the carboxylic acid group is automatically the first carbon of the chain

Esters

- Esters are one of the more challenging compounds to name
- Their name is based on the original alcohol and carboxylic acid that they were prepared from

- Ester names are confusing because the name is written backwards from the way the structure is drawn

- Methyl methanoate
  - The alcohol portion of the molecule contains the C–O single bond and is coloured green

bond and is coloured purple
        - The is one carbon, so this gives the methanoate part of the name
- Methyl ethanoate
    - The alcohol portion of the molecule contains the C–O single bond and is coloured green
        - There is one carbon, so this gives the methyl part of the name
    - The carboxylic acid portion contains the C=O double bond and is coloured purple
        - The are two carbons, so this gives the ethanoate part of the name
- Ethyl ethanoate
    - The alcohol portion of the molecule contains the C–O single bond and is coloured green
        - There are two carbons, so this gives the ethyl part of the name
    - The carboxylic acid portion contains the C=O double bond and is coloured purple
        - The are two carbons, so this gives the ethanoate part of the name

Examiner Tips and Tricks

Extended tier students should be able to draw the structural and displayed formulae for all of the compounds written above.

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